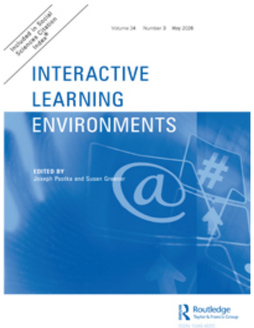

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EDITORIAL

Six global frameworks for human-centred AI literacy and competency: comparative analysis and a way forward

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Introduction

As artificial intelligence (AI) reshapes economies, education systems, and workplaces worldwide, governments and international bodies have rushed to define what citizens need to know about AI. The result is a growing collection of AI literacy and competency frameworks (Chiu, 2025), each tailored to specific audiences and national priorities. This editorial examines six major official frameworks from UNESCO, the OECD/European Commission, Australia, China, the United Kingdom and the United States. The analysis discusses both a global consensus on fundamental principles and notable differences in their implementation across regions. It highlights how the Human-Centric Artificial Intelligence Pedagogy (HCAP) framework (Chiu, 2026) can help educators develop these competencies in interactive learning environments.

The major frameworks and their core focus

UNESCO's AI Competency Framework for Students outlines competencies across four dimensions: human-centred mindset, AI ethics, AI techniques and applications, and AI system design. This framework structured competencies in three progression levels, i.e. Understand, Apply, and Create. It prepares students as responsible AI co-creators and critical citizens emphasizing ethical judgment, lifelong learning, and inclusive, sustainable design (UNSECO, 2024). The OECD-European Commission Joint Framework focuses on primary and secondary students, emphasizing cross-curricular integration and aiming to equip students with the knowledge, skills, and attitudes to understand, evaluate, and use AI ethically (OECD, 2025). Australia's Framework for Generative AI in Schools provides national guidance for ethical, responsible AI use in education. Endorsed by Education Ministers, it contains 6 Principles and 25 Guiding Statements across Teaching, Wellbeing, Transparency, Fairness, Accountability, and Privacy. The framework aims to enhance learning outcomes while protecting student data, ensuring equity, and maintaining human agency in decision-making (Australian Department of Education, 2023). China's Framework provides a comprehensive national K-12 curriculum guide with grade-specific objectives across primary, junior secondary, and senior secondary stages, adopting a "four-in-one" model combining knowledge, skills, thinking, and values within a national strategy for talent development (China's Cultivating Future-Oriented Innovative Talent, 2025). The United Kingdom's AI Skills Tools Package targets workforce development with three resources: a role-based AI Skills Framework (technical, ethical, non-technical skills across job levels), an Adoption Pathway Model, and an Employer Checklist for assessing readiness and planning inclusive upskilling (England, 2025). The United States Department of Labor Framework targets workforce development, offering five foundational content areas. They are Understand AI Principles, Explore AI Uses, Direct AI Effectively, Evaluate AI Outputs, and Use AI Responsibly, alongside seven principles for effective training delivery (USA training and employment notice no., 2026).

Core similarities across all frameworks

All six frameworks share fundamental principles that reflect a maturing international consensus on what AI literacy entails. Ethics is universally central; every framework explicitly addresses bias detection, data privacy,

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fairness, transparency, and responsible use as non-negotiable components. The US framework dedicates an entire content area to "Use AI Responsibly," while Australia's 25 Guiding Statements include robust protections for data and Indigenous Cultural and Intellectual Property. UNESCO's framework positions AI ethics as one of its four core dimensions alongside human-centred mindset, techniques, and system design.

Human-centricity runs through every document, emphasizing that AI should augment rather than replace human judgment. Australia's framework explicitly requires that "teachers and school leaders retain control of decision making and remain accountable." The UK framework stresses that "individuals in strategic roles remain grounded in core competencies." UNESCO's student framework cultivates "responsible AI co-creators and critical citizens" who exercise ethical judgment.

All frameworks move beyond technical skills, defining literacy as encompassing critical thinking, societal awareness, values, and the ability to evaluate AI outputs critically. The US framework's Evaluate AI Outputs domain requires workers to verify factual accuracy and apply human judgment. China's Values domain focuses on helping students understand AI's role in society and think carefully about both its benefits and possible harms. Each framework recognizes staged development. UNESCO structures competencies across Understand, Apply, and Create progression levels. China provides grade-specific objectives across primary, junior secondary, and senior secondary stages. The UK framework aligns skills with entry, mid, and managerial job levels in a cumulative structure.

Key differences

The frameworks diverge most significantly in whom they are designed to serve. For example, UNESCO with a global, competency-based model for responsible citizenship, and China with a nationally mandated K-12 curriculum for talent development. The OECD also addresses primary and secondary students but emphasizes cross-curricular integration across subjects like mathematics, history, and computer science rather than standalone instruction. In contrast, the US Department of Labor and UK frameworks are workforce centric. The US framework targets workers, job seekers, employers, and training providers with five foundational content areas applicable across industries. The UK provides a comprehensive skills tools package with a role-based framework, adoption pathway model, and employer checklist for assessing readiness and planning upskilling. Australia's framework addresses the entire school community – students, teachers, school leaders, support staff, parents, and policymakers – with specific guidance for generative AI use in school-based education.

The frameworks reflect fundamentally different structural natures. UNESCO is aspirational and formative, providing a global reference for countries to develop their own curricula rather than a prescriptive teaching tool. China is prescriptive and strategic, functioning as a policy instrument embedded within national education strategy. The OECD is integrative and assessable, designed for cross-curricular implementation. The US framework is instrumental and vocational and is a practical toolkit for workforce development with seven delivery principles including experiential learning, contextualization, and pathways for continued learning. The UK framework is operational and diagnostic with three interconnected resources that function as practical instruments for organizations to assess readiness, identify skills gaps, and plan upskilling activities. Australia's framework is protective and principled functioning as a safeguarding instrument for the entire school community.

HCAP helps educators connect the six major AI literacy frameworks

It is necessary to discuss how to implement the six frameworks (UNESCO, OECD, China, USA, UK, Australia) from a perspective of teacher capacity. While the six frameworks define the "what" AI literacy should look like for different populations, HCAP provides the "how" for educators to cultivate these competencies in learning environments (Chiu, 2026). HCAP provides a specialized teacher-focused pedagogical model that complements and operationalizes the six major AI literacy frameworks. It comprises five interconnected domains: I-TK (technical mastery of AI systems), I-CK (critical evaluation of AI content), I-PK (designing AI-enhanced learning), HAIC-K (structuring human-AI collaboration), and Ethics-K (ensuring responsible, equitable, and ethical AI use in education).

The HCAP framework provides the pedagogical "how" that operates the six frameworks' "what". Its five domains connect directly to framework components: I-TK (prompt engineering, system auditing) aligns with the US framework's Understand AI Principles and UK's technical skills. I-CK (source triangulation, bias detection) connects to Australia's Fairness principle and US's "Evaluate AI Outputs". I-PK (assessment redesign) operationalizes UNESCO's teacher competencies and Australia's learning design principles. HAIC-K (role allocation, critical dialogue) addresses the UK and US emphasis on human-AI collaboration. Ethics-K (equity auditing, well-being) permeates all frameworks' ethical requirements, i.e. Australia's Accountability, US's responsible use, and UK's ethical skills domains. Overall, the six frameworks establish the what of AI literacy across different populations, for example, students (UNESCO, OECD, China), workers (US, UK), and school communities (Australia). HCAP provides the how for educators, i.e. pedagogical knowledge and skills, to cultivate these literacies in learning environments. Together, they form a comprehensive ecosystem: frameworks define destination competencies; HCAP charts the teaching journey to reach them.

A way forward

This editorial discusses the six frameworks that define what AI literacy means for students, workers, and communities. These frameworks require a deeply human-centric approach to teaching. This is the way forward: empowering educators who champion critical thinking that places human values at the heart of technological progress, who nurture creativity that celebrates our unique human spirit, and who cultivate ethical reasoning that ensures AI serves humanity's deepest aspirations.

Looking ahead, it is vital to foster continued research exploring innovative strategies for implementing these frameworks in diverse educational contexts. By deepening our understanding of how teacher capacity-building can be scaled and adapted globally, scholars and practitioners will help ensure AI literacy is accessible and meaningful for all learners. Collaborative, interdisciplinary inquiry can reveal best practices and identify new challenges, laying the groundwork for more equitable, ethical, and impactful AI integration in education.

Disclosure statement

No potential conflict of interest was reported by the authors.

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